LAB-IV TASK 1 lab Report

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Section B

Title:

Implementation of Switch Statement for Days of the Week

Objective:

To develop a C++ program that takes a number (1–7) as input and displays the corresponding day using a switch statement.

Theory:

The `switch` statement is a multi-way decision-making structure used when there are multiple possible values for a single variable.

Syntax:

```
switch(expression)
{
    case value1:
        // code
        break;
    case value2:
        // code
        break;
    default:
        // code
}
```

- The `expression` is evaluated once.
- The matching `case` is executed.
- The `break` statement prevents execution of further cases.
- The `default` case handles invalid input.

Program (with comments):

```
#include <iostream>   // Input-output library
using namespace std;  // Standard namespace

int main()
{
    int day; // Variable to store day number

    cout << "Enter day number (1-7): "; // Prompt user
    cin >> day; // Take input

    switch(day) // Switch statement to check day
    {
        case 1:
            cout << "Monday" << endl; // If input is 1
            break; // Stop execution

        case 2:
            cout << "Tuesday" << endl;
            break;

        case 3:
            cout << "Wednesday" << endl;
            break;

        case 4:
            cout << "Thursday" << endl;
            break;

        case 5:
            cout << "Friday" << endl;
            break;

        case 6:
            cout << "Saturday" << endl;
            break;

        case 7:
            cout << "Sunday" << endl;
            break;

        default:
            cout << "Invalid input!" << endl; // If input is not 1-7
    }

    return 0; // End program
}
```

Detailed Explanation:

The program begins by declaring an integer variable `day`. The user inputs a number between 1 and 7.

The `switch` statement checks the value:

- Each `case` corresponds to a specific day.
- When a match is found, the respective day name is printed.
- The `break` statement ensures that only the matched case executes.
- If the input does not match any case, the `default` block runs.

Output (Example):

Enter day number (1-7): 3
Wednesday

Conclusion:

The switch statement simplifies multi-condition checking and makes code more readable compared to multiple if-else statements.